DGMS 2019 • SECTION I: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

Question 73: Groundwater Total Dissolved Solids (TDS) & Electrical Conductance (EC)

Aqueous Electrolytic Conduction, Units of Specific Conductance & Temperature Coefficients

OFFICIAL EXAMINATION QUESTION

Which of the following statements regarding the Total Dissolved Solids (TDS) and Electrical Conductance (EC) of the groundwater are correct?

1. A rapid determination of the TDS can be made by measuring the EC of the groundwater sample.

2. The specific conductance is measured in microsiemens/cm.

3. The specific conductance decreases with increase in temperature.

Select the correct answer using the codes given below:

(a) 2 and 3 only

(b) 1 and 3 only

(c) **1 and 2 only [CORRECT]**

(d) 1, 2 and 3

VERIFIED ANSWER KEY

Correct Option: (c) 1 and 2 only — Statement 1 is true because electrical conductance is directly proportional to ion concentration ($\text{TDS} \approx 0.55\text{--}0.75 \times \text{EC}$). Statement 2 is true because the standard international unit of specific electrical conductance is microsiemens per centimeter ($\mu\text{S/cm}$ at 25°C). Statement 3 is false because in aqueous electrolytic solutions,

warming decreases water viscosity and accelerates ionic mobility, causing specific conductance to **increase** by approximately 2% per $^\circ\text{C}$.

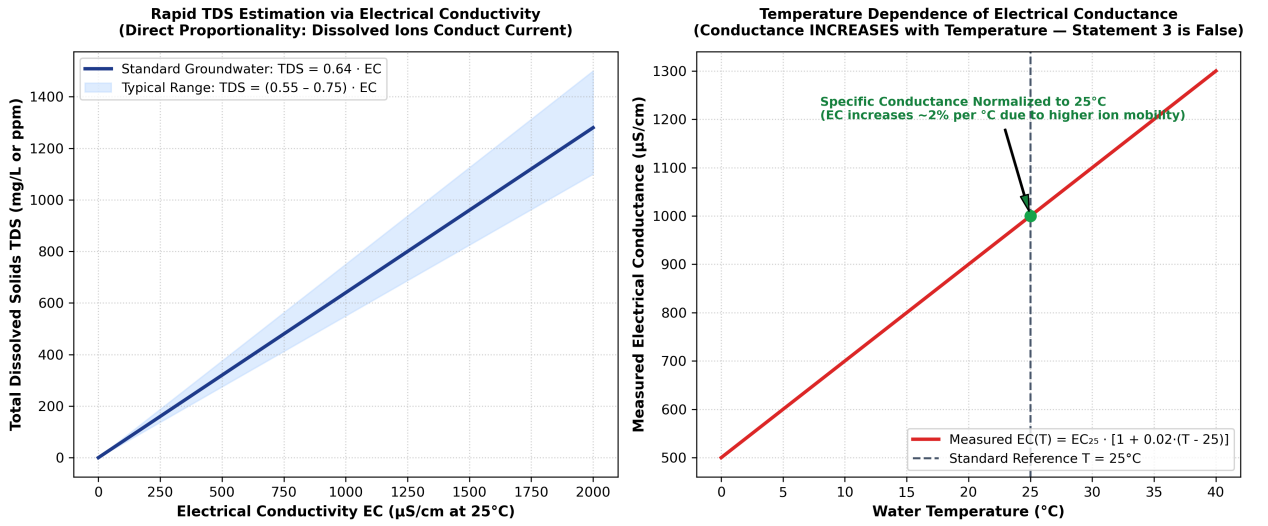


Figure 73.1 | Thermodynamics of Electrical Conductivity in Groundwater: (Left) Linear empirical correlation between Electrical Conductivity (EC in $\mu\text{S/cm}$) and Total Dissolved Solids (TDS in mg/L). (Right) Measured electrical conductance as a function of sample temperature; conductance rises steadily (~2% per $^\circ\text{C}$) because lower water viscosity promotes faster ionic migration, requiring standard reporting at 25°C.

1. Statement-by-Statement Diagnostic Breakdown

Statement 1: "Rapid determination of TDS via EC measurement" — CORRECT

Pure deionized water is an insulator. The electrical conductivity of groundwater arises exclusively from the presence of dissolved, dissociated ionic solutes (cations: $\mathrm{Na^+, Ca^{2+}, Mg^{2+}, K^+}$; anions: $\mathrm{HCO_3^-, Cl^-, SO_4^{2-}, NO_3^-}$).

- Gravimetric laboratory determination of TDS requires evaporating a filtered water sample to dryness at 180°C and weighing the residual salts, which takes hours.
- In contrast, a portable field conductivity meter instantly measures electrical current between platinum electrodes. In natural groundwater, TDS is related to EC by the well-established empirical linear equation: $$\text{TDS (mg/L)} = A \times \text{EC } (\mu\text{S/cm at } 25^\circ\text{C})$$ where conversion factor A typically ranges between **0.55 and 0.75** (average ≈ 0.64 for bicarbonate-chloride freshwaters, rising to 0.80 for sulfate-rich waters).

Statement 2: "Specific conductance is measured in microsiemens/cm" — CORRECT

Conductance (G) is the reciprocal of electrical resistance ($G = 1 / R$, unit: Siemens, S, formerly mho). Specific conductance (σ) normalizes conductance to a standard geometry (a 1 cm cube of solution):

- In the SI metric system and standard international hydrogeological practice (USGS, WHO, BIS), specific electrical conductance of water is expressed in **microsiemens per centimeter ($\mu\text{S/cm}$)** or millisiemens per meter (mS/m, where $1\ \mu\text{S/cm} = 0.1\ \text{mS/m}$).
- Fresh drinking water typically registers $100\text{ to }1000\ \mu\text{S/cm}$, while seawater registers $\sim 50,000\ \mu\text{S/cm}$.

Statement 3: "Specific conductance decreases with increase in temperature" — INCORRECT

Unlike metallic conductors (where electrical resistance increases with heat due to thermal lattice vibrations scattering electrons), aqueous electrolytic solutions behave in the exact opposite manner:

- In water, electric charge is transported by the physical migration of dissolved hydrated ions through liquid molecules.
- As temperature rises, thermal energy reduces water viscosity (μ) and weakens solvent-solute hydrogen bonds.
- Consequently, ionic drift velocity and mobility increase dramatically! The electrical conductivity of water **INCREASES** by approximately **$1.9\%\text{ to }2.2\%$ per degree Celsius increase**.
- To eliminate thermal variations, all specific conductance readings must be corrected to a standard reference temperature of **25°C** using the temperature compensation formula: $$\text{EC}_{25} = \frac{\text{EC}_T}{1 + \alpha (T - 25)}$$ where $\alpha \approx 0.02\ ^\circ\text{C}^{-1}$.

2. Groundwater Classification by Salinity & EC

Water Quality Class	Total Dissolved Solids (TDS, mg/L)	Specific Conductance (EC at 25°C, µS/cm)	Suitability & Use
Freshwater	< 1,000 mg/L	< 1,500 µS/cm	Potable drinking water; excellent for all irrigation.
Brackish Water	1,000 – 10,000 mg/L	1,500 – 15,000 µS/cm	

Water Quality Class	Total Dissolved Solids (TDS, mg/L)	Specific Conductance (EC at 25°C, µS/cm)	Suitability & Use
			Requires desalination (RO); tolerable only for salt-resistant crops.
Saline Water	10,000 – 35,000 mg/L	15,000 – 50,000 µS/cm	Unfit for human consumption or agriculture (coastal intrusion).
Brine	> 35,000 mg/L	> 50,000 µS/cm	Deep sedimentary basin formation fluids / evaporite pans.

3. Authoritative Textbook References

- **David Keith Todd & Larry W. Mays (2005)** *Groundwater Hydrology*, 3rd Ed., John Wiley & Sons, Chapter 8 ("Quality of Groundwater: Total Dissolved Solids & Specific Conductance"), pp. 325–340.
- **C. W. Fetter (2001)** *Applied Hydrogeology*, 4th Ed., Prentice Hall, Chapter 9 ("Groundwater Chemistry"), pp. 360–375.
- **John D. Hem (1985)** *Study and Interpretation of the Chemical Characteristics of Natural Water*, USGS Water-Supply Paper 2254, pp. 66–72.
- **H. M. Raghunath (2007)** *Ground Water*, 3rd Ed., New Age International Publishers, Chapter 7 ("Water Quality Standards").